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Importance of good/optimum water retention/drainage.	**Sources of errors during the experiment.**
➔ It reduces cases of drought/drought effects. ➔ It reduces leaching. ➔ It allows plants to absorb nutrients	➔ Poor folding of the filter paper. ➔ Using more than one filter paper. ➔ Failure to measure required amount of water/soil. ➔ Using wet apparatus. ➔ Poor timing technique.

Improving water retention in sandy soil.

- ❖ Application of organic manure, organic manure decomposes to form humus which helps in binding soil particles together thus improving on the water holding capacity of sand.
- ❖ Application of lime in the soil helps in binding soil particles together which improves the water holding capacity of sand soil.
- ❖ Marling sand with clay to improve sandy soil's ability to hold water.

Improving water drainage capacity of clay soil.

- ❖ Application of organic manure, organic manure decomposes to form humus which helps in binding soil particles together. This improves the porosity of clay thus improving its drainage capacity.
- ❖ Applying lime to the soil helps bind or aggregate clay particles. This improves the porosity of clay thus improving its drainage capacity.
- ❖ Marling clay with sand to improve the drainage capacity of clay soils.

Worked out item (about drainage).

Brandon has two plots of land C and D, in two different places. He plans to grow cassava in one of the plots but he doesn't know which of the plots is suitable for growing cassava. Specimens C and D are soil samples from plots C and D respectively.
(Available information shows that cassava grows well in soils that are well drained).

C is well dried sand soil. D is well dried loam soil.

Task.

Plan and carry out a suitable experiment and use the results to advise Brandon on which plot he should use to grow cassava. Advise him on how the plot that is not suitable for cassava growing can be improved to support cassava growing.

Possible response.

Aim.

An experiment to determine water drainage of soil sample C and D so as to advise Mr. Brandon which plot is suitable for growing cassava.

Hypothesis.

Soil sample C has a high drainage while soil sample D has a low drainage.

Variables.

Dependent variables. Volume of water collected.

Independent variables. Soil sample C and D.

Controlled variables. Volume of water used, volume of soil samples used and time taken

Materials and apparatus.

Water. Protective gears such as gloves. Labelling tape. Soil samples C and D. Filter papers. Filter funnels. Stop watching. 100ml measuring cylinders. A beaker.

Procedure.

1. Protective gears were put on for safety.
2. Materials and apparatus required to conduct the experiment were assembled.
3. Two measuring cylinders were labelled C and D using labelling tape.

4. A filter paper was folded and plugged into the filter funnel. The filter funnel was then placed onto measuring cylinder C.
5. 30cm^3 of soil sample C was measured using a beaker and poured into a filter funnel plugged with a filter paper on top of measuring cylinder C.
6. 35 cm^3 of water was measured using a beaker and poured into a filter funnel on top of measuring cylinder C and at the same time started the stop clock.
7. After 10 minutes, the funnel was removed from the measuring cylinder C and stopped the stop watch.
8. The amount of water collected in the measuring cylinder was recorded.
9. Steps 4 to 8 were repeated for specimen D.

Data presentation.

Soil Sample	Vol of soil used	Vol of water added	Vol of water collected
C	30cm^3	35cm^3	C>D
D	30 cm^3	35cm^3	

Data analysis and explanation.

High volume of water was collected from soil sample C meaning that sample C has a high water drainage and low volume of water was collected from soil sample D meaning that soil sample D has a low water drainage.

Soil sample C has big particles that are loosely packed which allows water to go through very fast while soil sample D has small particles that are closely packed which allows it to hold a lot of water.

Conclusion.

Soil sample C has a high drainage while soil sample D has a low drainage.

Recommendation.

I recommend Brandon to grow cassava in plot C since soil sample C which was obtained from pot C has a high and good drainage compared to soil sample D which was obtained from plot D.

Plot D can be improved to support cassava growing by: -

- Marling with sand soil to improve on the drainage of soil in plot D.
- Application of organic manure which decomposes to form humus that binds soil particles together thus improving on drainage.

Worked out item (retention)

Due to increasing demand for maize in the region, Mr. Abraham, wants to start large scale maize growing. He has identified two potential plots of land in the area on which to grow the maize. One is in Mpigi and one is in Wakiso. Soil experts have advised that soils from both places should be tested to decide the best land for his project. Soil samples were picked from the two areas for investigation; Sample **C** from Mpigi and sample **D** from Wakiso. *Maize requires good fertile soils with good/moderate water holding capacity to grow and produce high yields.*
C is well dried sand soil. D is well dried loam soil.
Task
Design and carry out a scientific investigation to determine which of the sample **C** or **D** is suitable for maize growing. Advise him on how the plot that is not suitable for maize growing can be improved to support maize growing.

Item 3.

Nabwiso, owns two plots of land in different parts of her village. She plans to grow groundnuts which grow best in soils that drain water quickly. To avoid poor yields, she wants to match the crop to the plot that best suits its water requirements. She collected soil samples from the two plots, labelled W1 and W2. She has approached you to help her.

Task.

Using the specimens W1 and W2 provided, design and carry out a scientific investigation and use your findings to advise Nabwiso. In your recommendation, also suggest how the other plot can be improved to support the growth of groundnuts.

Worked out item.

Nabirye owns two plots, one in **Busaano** and another in **Nyondo**. She plans to grow Irish potatoes in one of the two places but is not sure which plot is most suitable. However, it is known that Irish potatoes require well-aerated soils to allow proper tuber development and increased yield. Specimens **B** and **N** are soil samples obtained from Nabirye's two plots in Busaano and Nyondo respectively.
N is dry sand soil. B is dry loam soil.
Task
Plan and conduct a suitable experiment and use the results to recommend to Nabirye which of the two plots she should use for Irish potato growing.

Possible response

Aim.

An experiment to investigate the amount of soil air in soil sample B and N so as to advise Nabirye about which plot is suitable for Irish potato growing.

Hypothesis.

Soil sample B contains more air while soil sample N contains less air.

Variables.

Dependent variables. Volume of air in soil samples.

Independent variables. Soil samples B and N.

Controlled variables. Volume of water used, volume of soil sample used and time taken

Materials and apparatus.

Water. Protective gears such as gloves. Labelling tape. Soil samples N and B. Stopwatch. 100ml measuring cylinders. Stirring rod. Beakers.

Procedure.

a) Protective gears were put on for safety.
b) Materials and apparatus needed were assembled.
c) Two measuring cylinders were labelled N and B respectively using labelling tape.
d) 50cm^3 of soil sample N was measured using a beaker and poured into a measuring cylinder labelled N.
e) 50cm^3 of water was measured using a beaker and poured into a measuring cylinder labelled N.
f) The mixture in measuring cylinder N was stirred thoroughly using a stirring rod.
g) The mixture was left to stand for 10 minutes timed using a stop watch.
h) After 10 minutes, the stop watch was stopped and the final volume of the mixture was recorded.
i) Steps d) to h) were repeated for soil sample B.

Data presentation.

Sample.	Volume of soil sample used.	Volume of water used.	Expected total volume.	Final volume of the mixture.	Volume of air.
N	50cm^3	50cm^3	100cm^3	B<N	B>N
B	50cm^3	50cm^3	100cm^3		

Data analysis and explanation.

Soil sample B has more air because soil it has smaller particles which create numerous pores that keep a lot of air whereas soil sample N has less air because it has bigger particles with fewer pores that keep less air.

Conclusion.

Soil sample B has more air while soil sample N has less air.

Recommendation.

I advise Nabirye to grow Irish potatoes in the plot located in Busaano because soil sample B which was picked from it contains more air than soil sample N which was picked from Nyondo which encourages proper tuber growth and increased yield.

Trial items.

Item 6.

Kizza wants to plant cocoyam. Cocoyam grows best in well-aerated soils. He owns two plots of land but he is not sure of which plot is suitable for growing cocoyam. To avoid planting the wrong crop in the wrong plot, he brings the samples for testing. Two soil samples were collected from the two plots one in Kalama and on Kabuusu, labelled **A1** and **A2** respectively.

TASK

Using the specimens, A1 and A2 provided, carry out a scientific investigation and use your findings to advise Kizza on which plot is suitable for **cocoyam**. How can the other plot be improved to make it suitable for cocoyam growing?

Organic fertilizers.

These are fertilizers obtained from plant and animal remains or wastes. The most common organic fertilizers include farm yard manure, compost manure, green manure, and organic mulches.

The practical concepts assessed on organic fertilizers include;

i. Identification of organic fertilizers.
ii. Formation of organic fertilizers.

Common organic fertilizers.

1. Compost manure.

This manure is made from composting organic materials of plant origin. The materials include crop wastes, weeds, and domestic wastes. Compost manure can be made by two methods such as **pit method** and **heap method**. **Composting** involves decomposition of organic materials in pits or heaps by the help of microorganisms.

NB. Composting makes organic materials less bulky, it kills weed seeds and disease-causing organisms, and makes nutrients more available for absorption by plants.

2. Farm yard manure.

This is made from animal excreta and beddings. For example, cow dung, animal urine, chicken droppings, animal beddings and food remains of animals. These materials are mixed together and placed in one place to allow them to decompose. Farm yard manure should be made from liquid components and solid materials in a ratio of 3:1. The materials should be placed in an area with a concrete floor that is raised from the ground to reduce loss of plant nutrients.

3. Green manure.

This is manure formed by ploughing plants/crops grown specifically for crop improvement back into the soil. The plant should be ploughed into the soil before flowering to encourage maximum utilisation of plant nutrients.

Legumes are the most preferred plants grown for organic manure formation because of their ability to fix nitrogen in the soil and their high nitrogen content.

Advantages of organic fertilizers.	Disadvantages of organic fertilizers.
✓ They supply various plant nutrients. ✓ They add organic matter to the soil. ✓ They provide food to soil living organisms. ✓ They are cheap and easily accessible. ✓ They help maintain soil pH in optimum ranges. ✓ They do not require special skills to apply. ✓ They add humus to the soil which improves the water holding capacity of the soil. ✓ They form humus which improves soil structure by binding soil particles. ✓ They last longer in the soil.	✓ They may transfer pests, diseases and weeds into the garden. ✓ They are bulky hence difficult to handle and transport. ✓ Their application cannot be easily mechanized. ✓ If not properly decomposed, they can scorch plants (cause burning effects to plants.) ✓ Their specific nutrient content cannot be known. ✓ Their smell can be a source of diseases for farmers. ✓ They release available nutrients slowly.

Worked out item on fertilizer solubility.

S2 class of Mbeya High School are planning to establish a maize growing project in their school garden. For maximum productivity, the agriculture teacher has advised them to use inorganic fertilizers that should be applied at different stages of maize growth. Some have to be applied during planting while others have to be applied after crop establishment. Samples **H** and **J** represent the two fertilizers that they were advised to use. However, the students have been challenged in selecting the most appropriate fertilizer to use at different stages of maize growing.

H is Sulphate of Ammonia or Urea. J is SSP or TSP fertilizer.

Task.

a) Plan and carry out a scientific investigation using the two fertilizer samples and advise the S2 students accordingly.
b) What could be the negative effects of prolonged use of the two fertilizer samples for soil improvement.

Possible response.

Aim.

An experiment to investigate the solubility of fertilizer samples J and H to advise students of Mbeya High School on which fertilizers to use at different stages of plant growth.

Hypothesis.

Sample J is highly soluble in water while sample H is less soluble in water.

Variables.

Independent. Samples H and J.

Dependent. Solubility of the fertilizer samples.

Controlled. Volume of water used, Volume of fertilizer sample used.

Materials and apparatus. Samples H and J. Test tubes. Labelling tape. Water. 10cm^3 measuring cylinder.

Procedure.

1. Protective gears were put on for safety.
2. Materials and apparatus to be used were assembled.
3. Two test tubes were labelled H and J respectively using labelling tape.
4. 3cm^3 of fertilizer sample H was measured using a measuring cylinder and poured into a test tube labelled H.
5. 10cm^3 of water was measured using a measuring cylinder and poured into a test tube labelled H.
6. The mixture in test tube H was shaken vigorously for about 5 minutes measured using a stop watch.
7. After 5 minutes, the stop clock was stopped and the observations were recorded.
8. Steps 4 to 7 were repeated for fertilizer sample J

Data presentation

Sample.	Observation
H	All crystals dissolved completely in water to form a clear solution.
J	Some granules remain un dissolved at the bottom of the test tube

Data analysis and interpretation.

When water was added to a test tube containing Sample H followed by shaking, the sample completely dissolved meaning that H is highly soluble in water hence it is a Nitrogenous fertilizer.

When water was added to a test tube containing Sample J followed by shaking, the sample dissolved partially in water by leaving some residues at the bottom of the test tube. This means that sample J is a phosphatic fertilizer.

Conclusion

Specimen H is highly soluble while specimen J is less/partially soluble in water.

Recommendation.

Specimen H is highly soluble hence it is a Nitrogenous fertilizer. Therefore, it should be applied when the crops have established to maximize efficiency in nitrogen utilization in vegetative development and reducing cases of leaching.

Specimen J partially dissolves in water hence it is a phosphatic fertilizer. Therefore, it should be applied during planting. Since phosphorus is more essential in root development and sample H is not likely to over leach due to its low solubility.

Trial items.

Item 12.

Joan started a vegetable garden on her family land. To improve soil fertility. Her older brother placed three manure samples on a table and labelled them **O1, O2, and O3** He told her that each of these manures works differently on crops, and some require improvement before use. However, Joan is unable to

When the embryos of halves from specimens' B were stained with tetrazolium salt, all the embryos remained white in colour. This means that specimens' B are 0% viable.

Conclusion.

Specimens A are highly viable while sample B are less viable.

Recommendation.

I recommend the nursery operator to grow seed of the supplier from which specimen A were obtained. This is because seeds from sample A are highly viable hence, they can germinate easily.

Trial items.

Item 18.

Nanyonjo wants to plant beans at the beginning of the rainy season. She has two batches of seeds bought from different agro-input shops, labelled **V1** and **V2**. Because some farmers have recently complained about buying seeds that fail to sprout, she wants to test the viability of the two batches before planting so that she chooses the most reliable one.

TASK

Using the specimens **V1** and **V2** provided, carry out a scientific investigation and use your findings to advise her**.**

	Bark cracking and internal discoloration. Plant eventually dies.	Use of infected seedlings.	❖ Use resistant coffee varieties. ❖ Plant clean certified seedlings.
Ratoon stunting disease. **Cause:** bacteria	Stunted growth among plants. Thin and weak stalks Low yields	Use of contaminated cutting tools Planting infected sugar cane sets	❖ Use certified clean seed cane. ❖ Sterilize tools frequently before use. ❖ Destroy infected ratoon stools.
Cassava brown streak disease. **Cause:** Virus.	Leaf mottling and yellowing. Brown streaks on stems. Brown necrotic rotting inside roots.	Spread by whiteflies Use of infected cassava cuttings.	❖ Plant resistant/tolerant varieties. ❖ Rogue and destroy infected plants. ❖ Control whiteflies. ❖ Use clean certified cuttings.
Damping off. **Cause:** fungi.	Rotting of seedlings at soil level. Collapse of seedlings. Poor germination of seedlings.	Soil-borne fungi. Waterlogged soil condition. Use of contaminated nursery tools.	❖ Improve drainage. ❖ Avoid overcrowding. ❖ Use sterilised nursery soil. ❖ Treat seeds with fungicide.
Sigatoka / Banana leaf spot. **Cause:** fungus	Brown or black streaks/patches on leaves. Drying of leaves. Reduced bunch size.	Wind-borne spores. Rain splash which spreads spores. Infected leaf residues.	❖ Remove infected leaves. ❖ Apply recommended fungicides. ❖ Plant resistant varieties. ❖ Maintain good sanitation.
Banana wilt. **Cause:** bacteria	Yellowing and wilting of the plant. Bacterial ooze from cut stems. Premature fruit ripening. Pseudo stem splitting.	Contaminated farm tools. Insects spreading infected sap. Use of infected planting materials.	❖ Uproot infected plants immediately. ❖ Sterilize tools after every cut. ❖ Remove male buds to reduce insect spread. ❖ Use clean certified planting materials.

General control measures of crop diseases.

- ☒ Planting of resistant varieties.
- ☒ Crop rotation to disrupt the life cycle of disease-causing organisms.
- ☒ Uprooting and destroying infected plants.
- ☒ Carrying out closed season (A period when growing of certain crops is not done to break the life cycle of pests and pathogens)
- ☒ Applying recommended chemicals for some diseases.
- ☒ Use of health planting materials during planting.
- ☒ Destruction of crop residues.
- ☒ Disinfecting garden tools before and after use.
- ☒ Vector control for certain diseases by spraying with recommended chemicals.

Trial items.

Item 23.

Mr. Walusimbi, started clearing land for the next planting season. He observed that several weeds had invaded his fields, making land preparation costly and time-consuming. He uprooted samples of the weeds and brought them to the nearby school for assistance. He labelled the weeds **W1, W2, W3, W4, and W5**.

TASK

Observe specimens **W1 – W5**, do the following:

a) Classify each weed specimen by lifespan. Give a reason for the classification of each weed specimen.
b) Give the adaptations that make each weed difficult to control.
c) Suggest methods that can be used to control each weed specimen.
d) State the mode of propagation of each weed specimen.

	They help in maintaining body pH in optimum ranges (sodium, potassium and chlorine) They help in proper functioning of the nervous system (sodium, potassium) They help in formation of haemoglobin and transportation of oxygen in the body (iron). They help in formation of hormones like thyroid hormone (iodine).	
Lipids.	They provide energy to the body during periods of starvation. They are used during insulation to regulate body temperatures. They are used in cell membrane formation. They help store fat soluble vitamins like A, D, E and K. They help form hormones, mainly steroid hormones.	Oil feeds like cotton seed cake and groundnut seed cake
Vitamins.	They help boost the body's immunity (vitamin E, C, A). They help in maintaining sight for animals (vitamin A). They help in proper functioning of the nervous system (Vitamin B1, B2, B3, 12). They maintain normal body metabolism. They help in blood clotting (vitamin K). They help in boosting animals' reproductive processes (vitamin E). They help in the formation of bones (vitamin D). They help in removing toxins from the body (vitamin C).	Grass and legume pastures. Vitamin supplement.
Water.	Water is required in temperature regulation. Water is required in milk formation. Water is required during excretion of body wastes. Water acts as a solvent for minerals in the body of animals. Water acts as a medium for transport in the body of animals.	Water given to animals. Water from body metabolism. Water inherent within the feeds. (water in the feeds)

NB.

Lack of the above nutrients may lead to ***nutritional deficiencies***. Therefore, the farmer should feed animals on balanced rations to supply all the required nutrients to avoid nutritional deficiencies.

Worked out item on food tests.

Paul is a dairy farmer in Masaka district who is experiencing reduction in milk yield from his dairy cows. Upon inquiry from district agricultural officer, he was advised to supplement natural pastures fed on by the cows with a food sample that has more nutrients. Three food samples F, G, and H were presented to him but he is uncertain of the best food sample to use.
F is maize bran, G is soya bean cake and H is maize bran mixed with soya bean cake in a ratio of 1:1.
Task.
Design and carry out a scientific investigation on food samples F, G, and H and use your findings to advise Paul on which particular food sample to supplement the natural pastures.

Possible response.

Aim.

An experiment to investigate the nutrient content of specimens F, G, and H in order to advise Paul on which specimen to use while feeding his dairy animals.

Hypothesis.

Specimen F contains proteins, specimen F contains both proteins and carbohydrates while specimen and specimen H contains only carbohydrates.

Variables.

Dependent variables. Colour change of the food test reagents.

Independent variables. Type of reagent used, type of food nutrient in F, G and H.

Controlled variables. Volume of food sample used. Volume of reagents used.

Materials and apparatus. Iodine solution. Dilute sodium hydroxide solution. Copper (II) sulphate solution. Test tubes. Test tube rack. Measuring cylinder. Dropper. Labelling tape.

Procedure.

1) Protective gears were worn for safety.
2) Materials and apparatus needed were assembled.
3) Three test tubes were labelled F, G, and H using labelling tape.
4) $10cm^3$ of water was measured using a measuring and added into test tubes F, G and H followed by shaking to create food solutions.
5) The following tests were carried out on each of the specimen solutions separately.

   Starch test.

   $1cm^3$ of a food solution was measured using a measuring cylinder and poured into a test tube.

   3 drops of iodine solution were added into a food solution using a dropper.

   The observations and results were recorded.

   Protein test.

   $1cm^3$ of a food solution was measured using a measuring cylinder and poured into a test tube.

   $1cm^3$ of sodium hydroxide was measured using a measuring cylinder and poured into the test tube with a food solution.

   Three drops of copper (ii) sulphate solution were added into the mixture and then then shaken

   The observations and results were recorded.

Data presentation/observations.

Test.	Specimen	Observation.
Starch test	F	The turbid solution turned to black solution.
	G	The turbid solution turned to a brown solution.
	H	The turbid solution turned to black solution.
Protein test.	F	The turbid solution turned to colourless solution and then to blue solution.
	G	The turbid solution turned to colourless solution and then to purple solution.
	H	The turbid solution turned to colourless solution and then to purple solution.

Data analysis.

For specimen F, the food solution turned to black after a starch test while it turned to a blue solution on a protein test. Meaning specimen F contains starch but lacks proteins.

For specimen G, the food solution turned to a brown solution on a starch test while it turned to a purple solution on carrying out the protein test. Meaning specimen G contains proteins but lacks starch.

For specimen H, the food solution turned to a black solution on carrying out a starch test while it turned purple on carrying out a protein test. This means that specimen H contains both carbohydrates and proteins.

Conclusion.

Specimen F contains only starch, specimen G contains only proteins while specimen H contains both carbohydrates and proteins.

Recommendation.

Paul should feed his animals on specimen H because the food sample which was obtained from that specimen confirmed that specimen H contains starch which is a carbohydrate and proteins which are very essential in provision of a balanced diet for better production among dairy animals.

Trial items

Item 41.

A dairy farmer, **Kisakye**, buys calf starter feeds from two different shops. She wants to choose the feed that contains more of the essential food nutrients needed for healthy calf growth. To guide her decision, she brings samples of the two feeds labelled **F** and **G** for testing.

Task.

Using the specimens, **F** and **G** provided, carry out a scientific investigation and use your findings to advise him.

Item 45.

Paul visited a feed store in his school uniform. Upon entering the store, he found a farmer who had bought ingredients for mixing animal feeds represented by specimens U, V, W, X, Y and Z for his mixed farm. Since Paul was putting on a school uniform, the farmer challenged him to suggest reasons as to why each of the above ingredients is included in a ration for animals which puzzled him. The farmer was advised to feed his animals on a ration containing 15% crude protein. According to the labels on packages, V contains 10% crude protein while W contains 25% crude protein. The farmer is supposed to mix 500kg of feed for his animals with 2% of X however he doesn't know how to go about it.

Task.

a) Observe the specimens carefully and advise Paul about his challenge.
b) How can the farmer overcome his challenge?

Sera rears her fish in a pond that has capacity of producing 2 tons of fish annually. After harvesting, he sells her fish immediately after getting it from the pond. However, recently the demand for fish has reduced in the market. This forced her to stay with the harvested fish on the farm expecting price increase soon. On the farm, fish are kept in small plastic tanks. Specimen K is a sample picked from Sera's fish on his farm.

Task.

a) Carefully observe specimen K and explain its condition.
b) Give recommendations to Sera on what can be done to maintain her business with consistent profits.

Item 81.

Mpiima owns a charcoal stall in Mpala village. Specimen K1 is one of the by products from his charcoal business. In his locality specimens K2 and K3 are available. Mpiima would like to make an alternative to firewood for cooking his food. However, he doesn't know how to achieve his goal.

Task

Observe the specimens and advise Mpiima on how he can achieve his goal.

May God Bless You.

HOLISTIC GUIDE TO AGRICULTURE THEORY

A Comprehensive Textbook for the Advanced Secondary Curriculum

By

Kiyimba Nasser

Agriculture Teacher & Author

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DEDICATION

This book is respectfully dedicated to my wife whose unwavering support, patience, and encouragement have sustained me throughout the journey of writing and research. Her understanding and sacrifice have provided the foundation upon which this work has been completed.

It is also dedicated to my daughter, **Namulondo Rahmah**, whose presence has been a source of inspiration, joy, and renewed purpose. May this work stand as a testament to the values of hard work, perseverance, and dedication that I wish to pass on to her.

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1 AGRICULTURAL BIOLOGY IN CROPS AND ANIMALS.

Competency: **The learner demonstrates understanding of cellular Biology principles to illustrate their role in supporting agricultural production**.

1.1 Cells and cell physiology

Class activity 1. Exploring cells and their role in agricultural production

Materials needed

- ✧ *Microscopes and prepared slides of plant and animal cells.*
- ✧ *Drawing books or notebooks, pens/pencils*
- ✧ *Textbooks, internet, and agricultural reference materials*
- ✧ *Chart papers, markers for group presentations*
- ✧ *Diagrams/models of prokaryotic and eukaryotic cells*

Procedure

a) ***Introduction to cells***

- ➢ *In pairs, research the meaning of a cell and record findings.*
- ➢ *Discuss why cells are considered the basic unit of life.*

b) ***Importance of cells***

- ➢ *In groups, brainstorm and research the importance of cells to living organisms. Relate these functions to how they support crop and animal productivity.*

c) ***Prokaryotic vs eukaryotic cells***

- ➢ *Research definitions and differences between prokaryotic and eukaryotic cells.*
- ➢ *Create comparison charts.*

d) ***Parts of the cell and their structures***

- ➢ *Using diagrams, identify cell parts.*
- ➢ *Record each part, describe its structure, and explain its function.*

e) ***Plant vs animal cells***

- ➢ *Groups research and list similarities and differences between plant and animal cells.*
- ➢ *Draw labeled diagrams of each cell type.*
- ➢ *Discuss unique structures of each cell.*

f) ***Role of plant and animal cells in agricultural production***

- ➢ *Research how plant cells contribute to crop productivity.*
- ➢ *Research how animal cells contribute to livestock productivity.*
- ➢ *Prepare real-life agricultural examples for presentation.*

g) ***Reporting and presentation***

- ➢ *Each group prepares posters, charts, or PowerPoint slides summarizing findings.*
- ➢ *Present to the class with diagrams and explanations.*

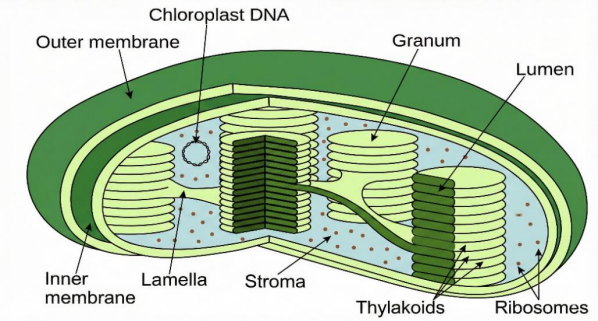
Cytology

Cytology is a branch of biology dedicated to the extensive study of cells. Cytologists use specialized instruments such as microscopes, staining techniques, and molecular tools to closely analyze and investigate cellular mechanisms, providing detailed insight into how living organisms function at the cellular level.

Reasons for studying cytology:

1. **Fundamental understanding of life:** Since all living organisms are composed of cells, studying cytology helps us understand essential biological processes such as growth, reproduction, energy use, and response to stimuli.
2. **Medical and veterinary applications:** Cytology is crucial for diagnosing various diseases by observing cellular abnormalities. It supports pathology labs in detecting cancers, infections, and other disorders, enabling accurate and early medical interventions in both human and animal health.
3. **Advancement in agriculture:** Detailed cytological knowledge aids agricultural scientists in developing genetically

Figure 1:5-Structure of the chloroplast.



Functions of the chloroplast.

- ⇒ Photosynthesis, it converts light energy into chemical energy (glucose).
- ⇒ Pigment storage, holds chlorophyll and other pigments for light absorption.
- ⇒ Production of oxygen, releases oxygen as a by-product of photosynthesis.
- ⇒ Storage of starch, stores starch grains formed from photosynthesis.
- ⇒ Self-replication, contains its own DNA and ribosomes, enabling it to produce some of its proteins and replicate independently.

c) **Mitochondria.**

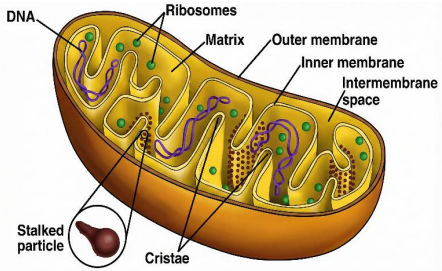
The ***mitochondrion*** (plural: mitochondria) is a double-membrane, bean shaped or oval-shaped organelle found in both plant and animal cells. It is often called the *"powerhouse of the cell"* because it produces energy in the form of ATP through cellular respiration. These are found within the cytoplasm of all eukaryotic cells. Though in some cells like in the red blood cells, they may be absent.

Structure of the mitochondria.

They are double membraned. The outer of which controls the entry and exit of chemicals. The inner one is folded inwards to give rise to extensions called cristae. The cristae increase the surface area on which the respiratory processes take place.

The surfaces of the cristae have stalked granules along their length. The remainder of the mitochondrion is the matrix which is a semi liquid material containing proteins, lipids and traces of DNA.

Figure 1:6-Structure of the mitochondrion



Function of the mitochondria.

- ♦ It produces energy in form of ATP molecules through aerobic respiration, powering cellular functions.
- ♦ It stores enzymes, DNA, and ribosomes for protein synthesis and self-replication.
- ♦ They can carry out self-replication due to presence of circular DNA.

d) Vacuoles.

A vacuole is a membrane-bound sac found in the cytoplasm of cells. Its structure varies between plant and animal cells:

Structure of a cell vacuole.

In plant cells, there is usually a single, large central vacuole that can occupy up to 90% of the cell's volume. It is surrounded by a special membrane called the **tonoplast**, which controls movement of materials in and out of the vacuole. The vacuole is filled with **cell sap**, a watery

agriculture mainly through photosynthesis, storage, and structural functions, while animal cells contribute through production of proteins, energy reserves, and reproduction. Together, they ensure the availability of crops, livestock, and related products that sustain human life and industry.

Roles of plant cells in agriculture

- Plant cells carry out photosynthesis, providing food (carbohydrates) that form the basis of the human and livestock diet.
- They store starch and sugars in vacuoles, which become harvestable products such as grains, tubers, and fruits.
- They synthesize plant proteins and oils, which supply nutrition in crops like beans, groundnuts, and sunflower.
- Plant cell walls rich in cellulose provide fiber and raw materials for textiles (cotton), paper, and ropes.
- Plant cells produce secondary metabolites (e.g., alkaloids, tannins, essential oils) used in medicine, pest control, and food processing.
- In crops, specialized plant cells (like xylem and phloem) support transport of water, nutrients, and food, ensuring growth and yields.
- Plant cells store pigments, vitamins, and minerals that enhance nutritional value of agricultural produce.
- They form protective tissues (cuticle, bark) that reduce losses from pests, diseases, and drought, contributing to crop resilience.

Roles of animal cells in agriculture

- Animal cells synthesize proteins, fats, and carbohydrates, which are harvested as meat, milk, eggs, and other animal products.
- Muscle cells in animals store energy and contract, providing high-quality protein (meat) for human consumption.
- Mammary gland cells produce milk, which is a vital agricultural product for nutrition and processing into dairy products.
- Reproductive cells (sperm and ova) enable animal breeding, supporting livestock production and genetic improvement.
- Blood cells in livestock transport nutrients and oxygen, maintaining animal health and productivity.
- Animal immune cells defend against disease, reducing mortality and ensuring reliable supply of animal products.
- Fat-storing cells (adipocytes) provide energy reserves and produce by-products like lard, tallow, and oils used in food and industry.
- Animal cells also produce fibers and hides—wool from sheep skin cells, silk from silkworm gland cells, and leather from cow skin.

Trial items 1:1-About the Cell.

Item one

Daniel recently joined a science club at his school. During one session, the teacher explained that a cell can be compared to a well-organized city, where different structures perform specific roles to keep the city alive and functioning.

Daniel is asked to imagine that problems have arisen in the "cell city," and he must correctly identify which organelle is responsible for each role in order to restore normal functioning.

He is given the following city roles and must decide which cell organelle best fits each role.

City roles

- ✓ The city headquarters which stores information and controls all activities.
- ✓ The power generation station which releases energy for all operations.
- ✓ The production workshops which manufacture protein materials for the city.
- ✓ The assembly and transport 1line which modifies and transports proteins and other materials within the city.
- ✓ The packaging and distribution Centre which packages, labels, and exports products.

- ✓ The Waste and Recycling Unit – digests worn-out materials and harmful substances.
- ✓ The storage reservoir which stores water, food, and waste materials.
- ✓ The security gate which regulates entry and exit of materials.
- ✓ The city wall which provides strength, protection, and shape.
- ✓ The solar power plant which manufactures food using sunlight.
- ✓ The city transport medium which holds all structures in place and allows chemical activities to occur.
- ✓ The internal framework which maintains shape of the city.

Task

a) Identify the cell organelle responsible for each city role and give one reason for each choice.

b) Distinguish between plant cells and animal cells.

1.1.3 Microscopic drawings

Class activity 2. Observing and drawing cells from crops and animals

Materials needed

- *Light microscopes, glass slides, cover slips, dropper, iodine solution (for plant cells), methylene blue (for animal cells), onion bulb, cheek swab (cotton bud), distilled water, filter paper or tissue, pencil and paper for drawing.*

Procedure

a) *Prepare a thin peel of onion epidermis and place it on a clean glass slide.*

b) *Add a drop of iodine solution to stain the onion peel.*

c) *Cover the specimen with a cover slip and place it on the microscope stage.*

d) *Observe the onion cells under low and high-power objectives.*

e) *Draw the shape of the onion plant cells as seen under the microscope.*

f) *Prepare a cheek cell specimen by gently scraping inside the mouth with a cotton bud.*

g) *Smear the sample on a clean glass slide and add a drop of methylene blue stain.*

h) *Cover with a cover slip and place on the microscope stage.*

i) *Observe the animal cheek cells under low and high-power objectives.*

j) *Draw the shape of the animal cheek cells as seen under the microscope.*

Microscopic drawings

Microscopic drawings of cells are scientific illustrations made after observing cells under a microscope. They are not artistic sketches, but accurate, simplified representations that highlight the main structures seen during observation.

When studying plant and animal cells, microscopic drawings help learners to:

- Visualize the **shape and arrangement** of the cells.
- Identify the **organelles** (like nucleus, cytoplasm, chloroplasts, vacuoles, mitochondria).
- Compare the **differences and similarities** between plant and animal cells.

In practice, drawings of plant cells often show a regular, rectangular shape with a cell wall, chloroplasts, and a large central vacuole, while drawings of animal cells show an irregular or round shape, with no cell wall or chloroplasts but visible nucleus, cell membrane, Golgi apparatus among other organelles.

Microscopic drawings must follow **scientific drawing rules**, such as:

- ✓ Use a sharp pencil with neat, clear lines.
- ✓ No shading—only single lines.
- ✓ Label parts using straight lines (no arrows, no overlapping).
- ✓ Provide a title and magnification (e.g., "microscopic drawing of a plant cell").

Significance of understanding biochemical compounds in agriculture.

a) Understanding biochemical compounds helps farmers provide balanced animal and plant nutrition since carbohydrates, proteins, lipids, vitamins, minerals, and water are the building blocks of life.

b) Knowledge of biochemical compounds enables formulation of quality animal feeds and rations that enhance growth, milk production, egg laying, meat yield, and overall animal health.

c) Awareness of biochemical compounds in crops guides fertilizer application and soil management by targeting nutrients like nitrogen, phosphorus, and potassium that support photosynthesis and protein synthesis.

d) Understanding enzymes and nucleic acids supports biotechnology applications such as DNA extraction, genetic engineering, and crop or animal breeding for improved traits.

e) Knowledge of biochemical compounds helps in detecting, diagnosing, and correcting nutritional deficiencies in farm animals and crops, thereby preventing diseases and losses.

f) Biochemical knowledge is applied in value addition to agricultural products, for example in milk processing (proteins, fats, enzymes), brewing (fermentation), or fruit preservation (vitamins, sugars).

g) Knowledge of biochemical processes improves post-harvest handling and storage of produce by controlling factors like respiration, enzymatic activity, and water loss.

Trial items 1:4-About biochemical compounds in cells.

Item one.

A farmer keeps cattle, goats, poultry, and pigs on the same farm. He notices that animals fed on balanced rations grow faster, resist diseases better, and reproduce normally, while those fed on poor-quality feeds show slow growth, weak muscles, low energy, and poor productivity. A veterinary officer explains that the performance of farm animals depends largely on the biochemical compounds present in their body cells, which support different activities.

Task

Explain the relevance of different biochemical compounds in farm animals.

Item two.

During a school outreach program, learners observed several conditions affecting plants, animals, and humans in a community.

Some calves in the area showed swollen bellies and slow growth. On a nearby farm, goats had weak bones and often fractured their legs easily. Chickens kept in the same area laid eggs with soft shells.

In another household, a farmer complained that his cattle get tired easily due low levels of substances responsible for carrying oxygen in the blood and reduced energy in body. His goats fall sick easily.

Vegetable crops in some gardens appeared stunted with yellow leaves, while others wilted quickly. In stored seeds, some failed to germinate, and cells taken from them showed reduced cell division due to lack of compounds that promote it.

Experts explained that many of these conditions were linked to imbalances of biochemical compounds in cells.

Task

For each condition described in the scenario, identify the biochemical compound whose deficiency or malfunction is most likely responsible.

Item three

During an agriculture lesson, learners examined information about living cells taken from plants and animals. They were told that cells contain different biochemical compounds

The also observed that cells contain a large amount of water and that many reactions only occur when water is present.

Task

a) Identify the biochemical compounds in cells responsible for the following roles:
   - ❖ Providing energy to cells
   - ❖ Building and repairing body tissues
   - ❖ Forming cell membranes and storing energy

2 ANATOMY, MORPHOLOGY AND PHYSIOLOGY

Competency: The learner demonstrates the relationship between anatomy and physiology of animals for increased production.

Class activity 9. Examining the shape, form, and appearance of animals
Materials needed
Chosen animal (e.g., goat, cow, rabbit, chicken, sheep), observation sheet or notebook, measuring tape (optional), camera or drawing materials, chart paper or whiteboard for group presentation.
Procedure
a) Form groups and agree on an animal to examine based on goals, available resources, and capabilities.
b) Observe the general body shape of the animal.
c) Examine specific features such as head, body, limbs, tail, skin covering (fur, feathers, scales), and any unique traits.
d) Record observations in a structured format using drawings, notes, or photographs.
e) Discuss within the group and agree on the description of the animal's form and appearance.
f) Relate observations to the uses and productivity of the animal.
g) Present findings to the class in written, drawn, or oral form.

2.1 **Morphology of farm animals.**

Meaning of morphology.

Morphology refers to the study of the **external physical characteristics**, shape, structure, form, color, size, and appearance of animals. It focuses on visible traits that differentiate animals.

For example, the horns of cattle, feathers of chickens, ear length in rabbits, and body shape in goats.

Morphology is especially important in agriculture because it helps farmers easily identify good-quality animals by observing their external features.

2.1.1 Morphological features of different animals

Morphological features are the visible physical characteristics and external structures of animals that can be observed without dissection. These features vary from one group of animals to another and often reflect their adaptation to the environment, feeding habits, mode of locomotion, reproduction, and protection against predators or harsh conditions.

Different animals show distinct morphological features that make it easy to identify them and understand their way of life. For example:

- ✧ Cattle and other mammals have fur or hair covering the skin, limbs adapted for walking or running, external ears for sound detection, mammary glands in females for feeding young, and a tail for balance or chasing insects.
- ✧ Birds possess feathers that aid in flight and insulation, wings modified from forelimbs, beaks adapted to different feeding habits, and lightweight skeletons.
- ✧ Fish have streamlined bodies for swimming, fins for movement and balance, scales covering the body, and gills for breathing in water.

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3 LIVESTOCK MANAGEMENT AND REARING PRACTICES.

Competency: **The learner applies evidence-based livestock management and rearing practices to ensure optimal quality production**.

Class activity 12. Animal enterprise establishment.
Materials needed
Observation checklist, notebooks, pens, camera/phone (optional), protective clothing (gumboots, lab coat, gloves), transport arrangements to the community farm.
Procedure
a) Organize a visit to a selected community farm or animal enterprise.
b) Observe and record how the farmer identified the type of animal enterprise chosen.
c) Examine the feasibility considerations. Ask the farmer about capital requirements, availability of feed resources, market demand, common risks and diseases, and profitability of the enterprise.
d) Study how the farm site was selected.
e) Observe and sketch the farm layout design and organization. Sketch the arrangement of different facilities on the farm.
f) Inspect how fences were established to secure the enterprise. Note the type of fence, reasons for fencing and advantages/disadvantages of fencing method used.
g) Examine animal handling structures. Observe all the structures and record their design, materials used, and suitability.
h) Identify equipment and farm inputs acquired by the farmer. List all the tools, equipment, other inputs and their uses.
i) Inquire about recruitment and training of labor on the farm. Inquire about recruitment, training, roles of workers, supervision, wages, and challenges in managing labor.
j) Observe how pastures or feed resources are established and maintained. Note the type of pastures grown, fodder conservation methods, supplementary feeds, and feed storage.
k) Examine the process of procuring breeding stock (sources, criteria for selection).
l) If present, capture how a fish ponds were established, type of ponds used, advantages/disadvantages.
m) Document findings through notes, photos, and drawings.
n) Return to class and compile a detailed group report or presentation covering all the steps observed (use a computer).
Guiding questions
✓ *What lessons can be applied to starting a similar enterprise in your own community?*

3.1 Establishing a farm animal enterprise.

Setting up a farm animal enterprise is a complex process requiring proper planning, feasibility assessment, and systematic design. It involves several aspects such as;

- ✓ Identifying type of animal to rear.
- ✓ Feasibility study of the animal enterprise.
- ✓ Selection of site for establishing the enterprise.
- ✓ Designing farm lay out.
- ✓ Fence establishment.
- ✓ Construction of animal handling structures.
- ✓ Acquisition of equipment and farm inputs.
- ✓ Recruitment and training of labour.
- ✓ Pasture establishment.
- ✓ Acquisition of the breeding stock

3.1.1 Identifying the type of animal enterprise

Selecting an animal enterprise is the process of choosing the most suitable type of livestock activity to rear or keep for a specific purpose such as income generation, food production,

- To carry out identification, for ear tagging, tattooing, branding, or other id methods.
- For weighing and record-keeping**,** helps monitor growth, production, and performance.
- To sort or select animals, for breeding, sale, slaughter, or culling purposes.
- To clean or move animals**,** for cleaning animal shelters or moving animals to new pastures.

How rounding up is done

1) Plan the rounding up activity in advance**,** choose a suitable time (morning or evening when it's cooler). Make sure the handling area or crush is ready.
2) Use calm and gentle methods to collect the animals**,** avoid shouting or hitting animals as it causes panic or injury.
3) Use herding tools or animals**,** use sticks, herding dogs, ropes, or calling sounds (for poultry and goats).
4) Move animals slowly into a holding area**,** guide them to a kraal, pen, crush, or holding unit for easy handling.
5) Restrain the animals if needed**,** use ropes, halters, crates, or crush pens for safety during treatment.
6) Monitor/observe the animals for any required aspect and carry out the required activity as needed.

3.2.7 Restraining in farm animals

Restraining is the act of holding or limiting the movement of an animal to allow safe handling or treatment without injury to the animal or the handler.

Reasons for restraining farm animals

- To safely handle animals during procedures like, castration, vaccination, deworming, hoof trimming, ear tagging.
- To avoid injuries to animals and handlers.
- To carry out health checks on mouth, eyes, hooves, etc.
- To assist in milking or feeding without interruptions.
- To load or transport animals safely.
- To aid in grooming or cleaning.

Figure 3:5-Methods of restraining farm animals.

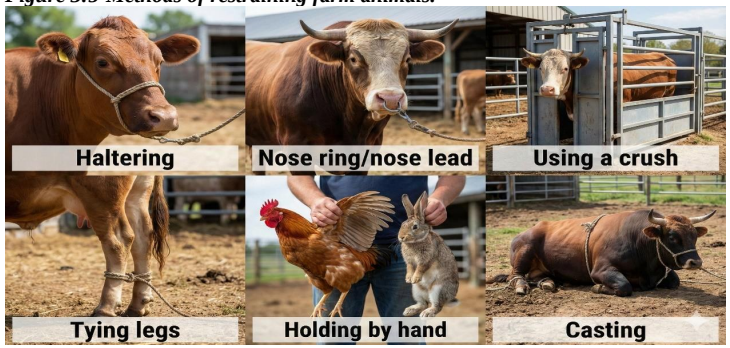


Table 3:6-Methods of restraining farm animals.

Method	Description	Common animals	When used
Haltering	Rope fitted on the head and around the nose	Cattle, goats	Leading, tying
Nose ring/nose lead	Metal ring or tong inserted in the nostrils	Bulls	Handling aggressive bulls
Using a crush	Narrow passage with side panels either V shaped or continuous crush.	Cattle, large goats	Vaccination, treatment

- Follow up with nutrition and hygiene to enhance immunity.
- Follow the manufacturer's instructions regarding the use of vaccines.
- Avoid exposing vaccines to high temperatures during transit.
- Do not vaccinate birds when they are under stress.
- Do not mix two vaccines together while administering
- Use distilled water in case you want to reconstitute the vaccine.
- All animals should be vaccinated at one time.
- Vaccination should be carried out in the cool hours of the day.
- The equipment to be used in the vaccination process should be disinfected.

Class activity 15. Capturing reproductive management practices in a farm animal enterprise

Materials needed

- *Observation checklist, notebooks, pens, protective clothing (gumboots, gloves, coats), camera/phone (optional), charts for reporting, textbooks for reference.*

Procedure

a) *Organize a visit to a selected community farm with breeding animals (e.g., cattle, goats, pigs, rabbits, or poultry).*

b) *Observe and record how **selection** of breeding stock is done. Note criteria used (health, productivity, body conformation, age, breed).*

c) *Examine how **breeding** is practiced on the farm. Record methods used (natural mating, artificial insemination, controlled mating) and procedures followed.*

d) *Study how **estrus** is detected and managed. Record signs of heat in animals, methods of detection, and management during the estrus cycle.*

e) *Observe how **natural mating and assisted reproductive technologies (ARTs)** are practiced. Record methods applied, procedures followed, and advantages/disadvantages of each.*

f) *Examine how **fertility management** is done. Record nutritional strategies, health care, body condition monitoring, and other measures used to maintain fertility.*

g) *Study cases of **infertility and sterility** in animals. Record common causes, methods of diagnosis, and how the farmer deals with such cases.*

h) *Observe how **flushing** (feeding animals with high-quality feed before mating) is practiced. Record feed types used, procedure followed, and expected benefits.*

i) *Document findings through notes, photos, and sketches.*

j) *Return to class and compile a detailed group report or presentation highlighting all reproductive management practices observed.*

3.2.20 Selection in farm animals.

In reality you cannot allow each animal to be a parent of the future generation. This is because some animals may have undesirable characteristics (traits) that you do not want to appear in other animals hence carrying out selection to choose the best animals for breeding.

Selection is the process of choosing animals that show desirable traits such as high milk yield, rapid growth, good fertility, or disease resistance to serve as parents for the next generation. By selecting the best individuals for breeding, farmers can gradually improve the overall characteristiscs, performance, and productivity of their herd or flock. This process may involve removing animals with undesirable traits from the breeding pool.

Selection aims to:

- Improve the productivity, health, and appearance of animals over generations.
- Pass on superior genes to offspring.
- Remove animals that are weak, diseased, infertile, or unproductive.

Class activity 16. Capturing reproductive and young animal management practices in a farm enterprise.

Materials needed

- *Observation checklist, notebooks, pens, protective clothing (gumboots, gloves, coats), camera/phone (optional), charts for reporting, textbooks for reference.*

Procedure

a) *Organize a visit to a selected farm where reproductive management and young animal rearing are practiced.*

b) *Observe how pregnancy in farm animals is diagnosed and managed. Record methods used, signs, care given, and feeding strategies during pregnancy.*

c) *Examine how drying off is practiced. Record when it is done, methods used, and reasons for drying off animals before parturition.*

d) *Observe how steaming up is carried out. Record the type of feeds provided, timing, and expected benefits for the animal and offspring.*

e) *Study how parturition is managed. Record signs of approaching parturition, procedures followed during birth, assistance given if needed, and care of mother immediately after birth.*

f) *Examine the management of young animals on the farm.*

g) *Observe how feeding of young animals is done. Record methods and types of feeds given.*

h) *Study record-keeping in farm animals. Record the type of records maintained, methods used and their importance.*

i) *Document findings through notes, sketches, and photos.*

j) *Return to class and compile a group presentation/report using a computer covering all practices observed.*

3.2.27 Pregnancy in farm animals.

Pregnancy in farm animals is the period between conception (fertilization of the ovum by sperm) and parturition (giving birth) during which the fertilized embryo develops into a fetus inside the uterus.

Pregnancy is a critical stage in livestock production as it determines herd replacement, offspring numbers, and productivity. Its length (gestation period) varies by species: e.g., cattle (about 280 days), goats (about 150 days), pigs (about 114 days), and rabbits (about 28 days). Successful pregnancy requires proper nutrition, good housing, careful handling, and regular health care to ensure the survival of both the mother and the offspring. Detecting pregnancy early allows farmers to manage animals properly by adjusting feeding, providing rest, and preparing for safe delivery.

Signs of pregnancy in farm animals.

After successful mating or insemination, it is important for farmers to confirm whether conception has occurred so that animals can be managed properly during gestation. The **signs of pregnancy in farm animals** are the observable physical and behavioral changes that indicate the presence of a developing fetus in the uterus.

These signs vary across species but generally include failure to return to heat, enlargement of the abdomen, udder and teat development, changes in appetite, and nesting or protective behaviors.

Early detection of pregnancy allows farmers to adjust feeding, provide special care, and prepare for safe delivery, thereby ensuring better survival and productivity of both the mother and offspring.

Class activity 17. Capturing fish, poultry, and rabbit management practices in community farms.

Materials needed

- *Observation checklist, notebooks, pens*
- *Protective clothing (gumboots, coats, gloves)*
- *Camera/phone (optional, with permission)*
- *Charts for reporting*
- *Textbooks/reference materials*

Procedure

Visit a mixed farm and carry out the following.

Fish management practices

a) *Observe pond site selection and construction.*
b) *Examine methods of pond preparation before stocking.*
c) *Observe stocking of fingerlings and record stocking densities.*
d) *Record feeds given to fish and feeding frequency.*
e) *Observe water quality management practices (aeration, cleaning, inlet/outlet control).*
f) *Record pond fertilization practices for plankton growth.*
g) *Examine pond weed and predator control.*
h) *Observe fish health management practices, including disease control.*
i) *Record growth monitoring and sampling methods.*
j) *Observe harvesting techniques used on the farm.*
k) *Record methods of handling and storage of harvested fish.*
l) *Observe marketing practices for fish in the community.*

Poultry management practice

a) *Observe chick brooding practices and heating methods.*
b) *Record housing design and ventilation systems.*
c) *Observe poultry stocking rates and spacing.*
d) *Record feeds given at different growth stages.*
e) *Observe water provision methods.*
f) *Examine vaccination schedules and health care.*
g) *Record parasite control practices (internal and external).*
h) *Observe egg collection, handling, and storage.*
i) *Record waste management and hygiene practices in poultry houses.*
j) *Observe record-keeping in poultry enterprises.*
k) *Record culling practices for unproductive birds.*
l) *Observe marketing of eggs and meat in the community.*

Rabbit management practices

a) *Observe rabbit hutch construction and arrangement.*
b) *Record stocking density and space requirements per rabbit.*
c) *Observe types of feeds provided and feeding frequency.*
d) *Record provision of clean drinking water.*
e) *Observe rabbit breeding management and mating systems.*
f) *Record methods of kindling (birth) preparation.*
g) *Observe weaning of young rabbits.*
h) *Record grooming and hygiene practices.*
i) *Observe disease control and vaccination if applicable.*
j) *Record waste disposal and hutch cleaning methods.*
k) *Observe record-keeping for breeding, health, and sales.*
l) *Record methods of marketing rabbits and rabbit products.*

Cross-cutting

✦ *What are the advantages and disadvantages of each enterprise?*

4 HARVESTING FARM ANIMALS AND ANIMAL PRODUCTS

Competency: The learner harvests optimal quality farm animals and animal products for the market.

Class activity 18. Observing the harvesting of farm animals and animal products.
Materials needed
• *Observation checklist, notebooks, pens/pencils* • *Protective clothing (gumboots, lab coats, gloves, masks)* • *Camera/phone (optional, with permission)* • *Chart papers/markers for reporting after the visit* • *Reference textbooks*
Procedure
a) Organize a field visit to a selected community farm (or farms) where milk, meat, fish, and poultry broilers are harvested.
b) Observe the harvesting of milk in dairy animals. Record preparation of the cow, milking methods (hand, machine), hygiene practices, handling, storage, and preservation of milk.
c) Observe the harvesting of beef animals (slaughter process). Record pre-slaughter handling, stunning, bleeding, flaying, evisceration, carcass inspection, meat storage, and hygiene practices.
d) Observe the harvesting of broiler chickens. Record catching and handling of birds, slaughtering methods, bleeding, de-feathering, evisceration, carcass grading, packaging, and storage.
e) Observe the harvesting of fish from ponds or cages. Record harvesting methods and handling.
f) Compare similarities and differences across the enterprises in terms of procedures, safety, hygiene, and market readiness.
g) Record challenges faced by farmers in harvesting animals and products.
h) Document all findings in notes, sketches, and photos.
i) Return to class and prepare a group report or presentation highlighting practices, procedures, and lessons learned.
Cross-cutting questions.
• *What occupational health and safety measures are observed during harvesting?*
• *What environmental considerations are taken into account during harvesting?*
• *What challenges are faced in harvesting animals and animal products?*
• *What lessons can be applied to improve harvesting practices in your community?*

4.1 Preparation for harvesting farm animals and animal products

Preparation for harvesting farm animals and their products refers to all the planned activities and management practices carried out before animals are slaughtered for meat or before products such as milk, eggs, wool, and honey are collected. The main aim is to ensure quality, safety and marketability of the final product while also safeguarding the welfare of the animal and the health of the consumer.

Reasons for preparation before harvesting animals and animal products

- ➔ **To ensure good quality of products,** preparation helps in producing safe and attractive animal products. For example, cleaning udders before milking gives clean milk free from dirt and bacteria, while fasting cattle before slaughter reduces gut content and results in cleaner meat.
- ➔ **To protect consumer health,** preparation reduces the risk of contamination and disease transmission. For example, veterinary inspection of animals before slaughter ensures diseased cattle are not

and wastewater must be treated or disposed of in safe ways that do not harm human health or the environment.

- ➢ *Pollution of land, water, and air must be avoided*, slaughter operations are required to manage effluents and by-products so that rivers, soil, and the atmosphere are not contaminated with harmful waste.

Importance of international regulations in harvesting farm animals

- ✧ **They protect animal welfare,** the rules ensure animals are transported, handled, and slaughtered humanely, which prevents unnecessary pain, suffering, and stress, for example stunning cattle before slaughter makes the process more humane.
- ✧ **They safeguard consumer health,** by enforcing drug withdrawal periods and strict veterinary inspections, these regulations prevent harmful residues and diseased products from reaching the market, for example milk from cows treated with antibiotics is withheld until safe.
- ✧ **They ensure food safety and quality,** by demanding hygiene in abattoirs and during product handling, the rules guarantee that meat, milk, eggs, and fish are clean, free from contamination, and safe for consumption.
- ✧ **They improve marketability and trade,** compliance with international standards allows farmers and countries to access local and global markets, for example Ugandan beef or poultry can be exported when it meets hygiene and welfare regulations.
- ✧ **They reduce economic losses,** regulations prevent carcass rejection, spoilage, and downgrading by enforcing proper handling, transport, and inspection of animals before and during slaughter.
- ✧ **They promote traceability and accountability,** tagging and documentation systems help track animal products from farm to market, making it possible to trace disease outbreaks and build trust with consumers.
- ✧ **They protect the environment;** by requiring proper waste disposal and pollution control, the regulations prevent contamination of rivers, land, and air from slaughterhouse waste like blood, feathers, and wastewater.
- ✧ **They build consumer confidence,** when people know that animal products are inspected, safe, and produced under humane conditions, they trust the farmers and are more willing to buy the products.
- ✧ **They promote sustainable farming,** by ensuring welfare, safety, and environmental care, the regulations support long-term livestock production without damaging public health or the environment.

Trial items 4:1-About harvesting animals and animal products.

Item one

Nakato keeps layers in a deep-litter poultry house. Egg collection is done once daily when other farm duties allow. She enters the poultry house quickly and moves around while the birds are still active.

Eggs are collected using bare hands and dropped into a plastic basin. Most eggs are laid in corners and on the litter.

Cracked and soiled eggs are mixed with other eggs during collection to simplify work. When eggs are difficult to reach, they are pulled out using a piece of wood. During collection, the basin is occasionally placed on the floor as she inspects litter.

Task.

Advise Nakato on how to improve egg harvesting from her poultry enterprise.

Item two.

Achan keeps dairy cattle under a semi-intensive system. On milking days, she milks whichever cows are present in the kraal at the time. Calves are allowed to remain with their mothers until milking begins.

5 VALUE ADDITION TO FARM ANIMAL AND ANIMAL PRODUCTS.

Competency: The learner adds value to the farm animal and to animal products for the market.

Class activity 19. Post-harvest handling and value addition of animal products

Materials needed

- *Observation checklist, notebooks, pens/pencils*
- *Protective clothing (gumboots, coats, gloves, masks, hairnets)*
- *Camera/phone (optional, with permission)*
- *Reference materials (textbooks, internet access if possible)*
- *Chart papers/markers for group presentations*

Procedure

a) *Organize a farm visit to observe post-harvest handling of animal products (milk, meat, eggs, fish).*

b) *At the farm, observe and record how products are handled immediately after harvest. Participate where possible.*

c) *Inquire from farmers, workers, or processors about value addition methods applied to each product.*

d) *Record and discuss the preservation methods observed or explained for various products such as milk, fish, meat, eggs. Participate in procedure of implementing these methods where possible.*

e) *Investigate and carry out procedures of preservation following step-by-step practices.*

f) *Observe, participate and record processing methods applied on the farm for products such as milk, fish, meat and eggs.*

g) *Inquire about the procedures of processing. What equipment, steps, and safety measures are followed.*

h) *Examine the market functions applied for processed products.*

i) *In groups, design a packaging, branding, labelling and advertising materials for animal products adhering to the local and international regulations. Use computer software such as Photoshop and publisher.*

j) *Document findings during the farm visit in notes, sketches, or photographs.*

k) *After the visit, form research groups to:*

- *Compare observed practices with recommended practices from textbooks and online sources.*
- *Research further on value addition, preservation, processing methods, and marketing of milk, meat, eggs, and fish.*
- *Develop **detailed group reports** that integrate farm visit observations and secondary research.*

k). Present findings in class through posters, PowerPoints, or charts showing:

- *Post-harvest handling procedures.*
- *Value addition methods and products.*
- *Preservation and processing procedures.*
- *Market functions applied.*

Cross-cutting

- ✦ *What occupational health, safety, and environmental protection measures are followed during handling and processing?*
- ✦ *What challenges are faced in post-harvest handling, preservation, processing, and marketing?*
- ✦ *What lessons can be applied in the community to improve value addition, preservation, processing, and marketing of animal products?*

6 ANIMAL PRODUCT MARKETING.

Competency: The learner markets the farm animal and animal products for profit

Class activity 20. Marketing of animals and animal products

Driving question

How can we research, analyze, and execute an effective marketing strategy that promotes animals or animal products in our community?

Materials needed

- *Market survey tools (questionnaires, interview guides, observation checklists)*
- *Access to local markets, farms, cooperatives, or traders*
- *Notebooks, pens/pencils, clipboards*
- *Computers/phones (for data analysis)*
- *Chart papers, markers, sample packaging materials*
- *Reference textbooks and internet (if available)*

Procedure

Phase 1: problem identification and planning

1. Form groups and select one animal or animal product such as milk, eggs, broiler meat, fish, rabbit meat, goat milk.

2. Brainstorm marketing challenges faced by this product in the community for example limited demand, poor storage, weak promotion and lack of processing.

3. Frame a research question for example "How can we improve the marketing of [chosen animal/product] in our community?"

Phase 2: market research

4. Design and administer survey tools such as questionnaires, interviews, observation checklists targeting farmers, traders, consumers, processors, and market leaders.

5. Collect data on:

- *Current demand and supply trends.*
- *Price variations and consumer preferences.*
- *Seasonality of demand and supply.*
- *Major competitors and substitute products.*

6. Document findings with tables, charts, and summaries.

Phase 3: market analysis

7. Analyze data to identify:

- ➢ *Trends such as rising demand, seasonal dips and popular products.*
- ➢ *Opportunities such as niche markets, value addition and untapped consumers.*
- ➢ *Challenges in aspects like transport, storage, competition and perishability.*

Phase 4: market channels investigation

8. Investigate available market channels for the chosen product,

- ✧ *Direct sales (farm-gate).*
- ✧ *Local markets.*
- ✧ *Wholesalers and retailers.*
- ✧ *Hotels, restaurants, schools, and institutions.*
- ✧ *Supermarkets and processors.*
- ✧ *Export markets (if applicable).*

9. Analyze advantages and disadvantages of each channel (cost, reach, profitability, risks).

Phase 5: marketing strategy design

10. Design a marketing strategy using the 4Ps (Product, Price, Place, Promotion) framework,

- ✦ *Product, define the product form (raw, processed, value-added).*
- ✦ *Price, choose a pricing strategy (competitive, premium, cost-plus).*
- ✦ *Place, select distribution channels.*
- ✦ *Promotion, develop tools for creating awareness such as radio adverts, posters and social media posts.*

11. Include aspects of branding, packaging, labeling, storage, and financing.

Phase 6: strategy execution (simulation/practice)

12. Prepare campaign materials such as posters, sample packed products, social media posts, radio

jingles, packaging designs.

13. Role-play or simulate a marketing campaign in class or at the community level.

14. Collect feedback from peers, teachers, or community members on the effectiveness of the campaign.

Phase 7: presentation and reporting

*15. Compile findings in a **project report**, including:*

- *Market research results.*
- *Trend and opportunity analysis.*
- *Market channel evaluation.*
- *Designed marketing strategy.*
- *Campaign execution and feedback.*

16. Present the report to the class with visuals such as charts, sample packed products, posters, and digital slides.

Guiding questions

- *What trends exist in the market for the chosen animal/product?*
- *What opportunities can be tapped to improve marketing of the product?*
- *What challenges are faced in marketing and how can they be solved?*
- *Which market channels are available and which are most effective?*
- *What marketing strategies can be applied to increase demand and profitability?*
- *How can the product be packaged, labeled, priced, promoted, and distributed effectively?*
- *What lessons can be applied to improve marketing of farm animals and products in your community?*

6.1 Marketing of animal products.

Marketing of animal products refers to all activities involved in the movement of animal-derived goods from the point of production to the final consumer. These activities include planning, processing, packaging, pricing, promotion, distribution, and sales of products obtained from livestock, poultry, fish, and other domesticated animals.

Examples of animal products.

a) Meat products**,** beef, pork, goat meat, mutton, poultry meat (chicken, duck, turkey), rabbit meat.
b) Milk and dairy products, fresh milk, cheese, yogurt, butter, ghee.
c) Eggs, chicken eggs, duck eggs, quail eggs.
d) Hides and skins, from cattle, goats, sheep, and pigs.
e) Wool and hair, from sheep, goats (cashmere), rabbits (angora).
f) By-products, bones, horns, hooves, blood meal, feathers, animal fat (tallow), manure for organic farming.

Importance of marketing animal products.

- ✧ Ensures income generation for farmers and producers.
- ✧ Encourages increased animal production through profitability.
- ✧ Facilitates availability of animal-based foods to consumers.
- ✧ Stimulates rural and urban economic development.
- ✧ Helps reduce post-harvest losses through timely disposal of perishable products.
- ✧ Supports value addition and agribusiness development.

6.1.1 Market research for animal products.

Market research is the systematic process of gathering, analyzing, and interpreting information related to a specific market, particularly consumer needs, preferences, behaviors, and trends. It provides factual data used to guide decision-making in marketing.

In animal product marketing, market research involves collecting and analyzing data on the demand, supply, pricing, competition, customer preferences, trends, and satisfaction for animal and animal-based products.

7 LAND POLICY.

Competency: The learner demonstrates an understanding of land rights, land tenure and land reform to ensure sustainable crop production.

Class activity 21. Exploring land reforms and land tenure systems in Uganda for sustainable agriculture

Materials needed

- *Textbooks, internet access, local government reports on land reforms*
- *Notebooks, pens/pencils.*
- *Sample letter-writing formats*
- *Chart papers and markers for role play scenarios*
- *Copies of Uganda's Constitution (land sections), Land Act (1998), and amendments*
- *Props for role play (optional: sticks for boundary marks, sample land titles, placards)*

Part i. Individual research and advocacy letter writing

Procedure

a) *Research and collect information on different land reforms in Uganda.*
b) *Investigate land tenure systems in Uganda.*
c) *Record the characteristics, advantages, and disadvantages of each tenure system.*
d) *Analyze how these land tenure systems influence sustainable crop production in the community.*
e) *Individually, write a letter addressed to local authorities such as the District Land Board, LC5, or Ministry of Lands advocating for land reforms that can promote sustainable crop production.*

Guiding questions for research

- *What land reforms have been implemented in Uganda since colonial times?*
- *What are the characteristics of land tenure systems in Uganda?*
- *What are the advantages and disadvantages of each land tenure system?*
- *How do land tenure systems affect access to land for farming in the community?*
- *What reforms are needed to ensure sustainable crop production?*
- *How can local authorities collaborate with communities to improve land management?*

Letter writing guidelines

- *Identify the land reform you are advocating for.*
- *Explain the issue or challenge in your community related to land tenure.*
- *Propose reforms or solutions that will promote sustainable crop production.*
- *Use persuasive and respectful language to advocate for change.*

Part ii. Group role play on land tenure issues.

Procedure

a) *Form groups and select a realistic land tenure scenario to role play, such as:*
    - *A dispute over land ownership between two families.*
    - *A conflict over unclear boundary marks.*
    - *A community struggling to access land for farming due to tenure restrictions.*
    - *A tenant farmer negotiating with a landlord for land use rights.*
b) *Assign roles such as farmers, landlords, tenants, local council leaders, elders, mediators, and government officials.*
c) *Develop a short script or outline to present the scenario.*
d) *Perform the role play in class, showing how stakeholders collaborate, communicate, and resolve the issue.*
e) *After the performance, discuss as a class the lessons learned about:*

8 SOIL SYSTEMS.

Competency: The learner demonstrates understanding of soil systems for crop productivity.

Class activity 22. Collecting and analyzing soil samples from the community

Materials needed

- *Soil auger/spade/hoe, clean polythene bags or containers, permanent marker*
- *Drying trays, mortar and pestle, sieve (2 mm mesh)*
- *Weighing balance, measuring cylinder, beakers, funnels, filter paper*
- *Oven or heat source, distilled water*
- *pH testing reagents and apparatus.*
- *Microscope and petri dishes.*
- *Crucibles, chemicals such as HCl and NaOH if available.*
- *Lab coats, gloves, notebooks, pens*

Procedure

A. Pre-analysis preparation

a) *Identify sampling sites in the community.*

b) *Identify 20 spots from the field for use during sampling and mark them with pegs.*

c) *Collect soil samples using an auger/spade from 15–20 cm depth.*

d) *Mix sub-samples to form a composite sample. Place samples in labeled polythene bags.*

e) *Transport samples to the laboratory or classroom carefully to avoid contamination.*

f) *Air-dry soil samples on trays, remove plant residues and stones, grind gently with a mortar and pestle, and sieve through a 2 mm mesh.*

B. Physical analysis. Analyze different properties as follows,

a) *Soil texture, use sedimentation, rolling method or feel method to determine sand, silt, and clay proportions.*

b) *Particle size distribution, use sieving/sedimentation method to separate soil particles into fractions.*

c) *Water holding capacity, use measuring cylinders, filters papers, stop watches and water to determine the water holding capacity of each sample.*

d) *Bulk density, weigh oven-dried soil of a known volume and calculate its bulk density.*

e) *Particle density, calculate the particle density of a soil sample.*

C. Chemical analysis.

a) *Soil pH, mix soil with distilled water test pH using universal indicator.*

b) *Nutrient analysis, test for nitrogen, phosphorus, and other nutrients if possible.*

c) *Record results and compare with recommended levels for crops.*

D. Biological analysis.

a) *Microorganisms count, investigate the presence of microorganisms in fresh soil sample.*

b) *Organic matter content, investigate the presence of organic matter in a soil sample using oxidation method or ignition method.*

15. Record results and interpret them.

E. Reporting.

a) *Compile group results into a report showing sampling sites, physical, chemical, and biological properties of soils.*

b) *Discuss implications of results on crop production and soil management.*

c) *Present findings to the class through charts, graphs, or posters.*

9 CROP PRODUCTION AND HUSBANDRY.

Competency: The learner applies evidence-based husbandry practices to ensure optimal quality production.

Establishing a crop enterprise

Class activity 24. Establishing a crop enterprise in the community.

Materials needed

- *Notebooks, pens/pencils, and clipboards*
- *Chart papers and markers for planning exercises*
- *Sample feasibility study templates*
- *Soil sampling tools (spade, auger, polythene bags)*
- *Access to land or a demonstration site (school garden/community farm)*
- *Reference materials on crop enterprises in Uganda*
- *Locally available farm inputs (seeds, fertilizers, tools)*

Procedure

Step 1: identifying the purpose of the enterprise.

a) *As a class, brainstorm possible reasons for starting a crop enterprise.*

b) *Discuss how community needs and available resources shape the purpose.*

c) *Record selected purpose for the enterprise.*

Step 2: acquiring land for establishing the enterprise

a) *Inquire about sources of land.*

b) *Evaluate challenges of land access using Uganda's land tenure systems.*

c) *Document how farmers in the community acquire land for crop enterprises.*

Step 3: selecting suitable crops

a) *Research crops suitable for the local agro-ecological conditions.*

b) *Consider factors such as rainfall, soil type, market demand, capital, and labor.*

c) *List crops that are most profitable and sustainable for the enterprise.*

Step 4: conducting a feasibility study for the crop enterprise

a) *Design a feasibility study outline covering,*

- *Market demand and availability*
- *Availability of inputs such as seeds, fertilizers, tools.*
- *Availability of labor and technical knowledge*
- *Financial requirements and profitability analysis.*
- *Risks such as climate, pests, diseases and market fluctuations.*

b) *Interview local farmers or extension workers for real-life data.*

c) *Summarize findings in a feasibility study report.*

Step 5: site selection.

a) *Visit potential sites and evaluate basing on,*

- *Soil fertility and type*
- *Water availability*
- *Accessibility including roads, markets and storage facilities.*
- *Topography and drainage*
- *Security and land tenure clarity*

b) *Choose the best site for the enterprise.*

Step 6: land preparation.

a) *Observe and participate in land clearing, ploughing, harrowing, and leveling.*

b) *Discuss advantages and disadvantages of manual vs mechanized land preparation.*

Step 7: soil testing and addition of soil amendments.

a) *Collect soil samples using standard procedures (composite sampling at 15–20 cm depth).*

b) *Send samples for laboratory analysis or conduct school-level tests.*

c) *Recommend soil amendments.*

d) *Apply soil amendments as per recommendations.*

Step 8: designing a farm layout for the crop enterprise.

a) *In groups, sketch different possible layouts showing:*

- *Crop plots (rotation and intercropping designs)*
- *Irrigation channels or water points*
- *Farm roads/paths*
- *Storage sheds or stores*
- *Fencing and windbreaks*

b) *Select the most efficient layout based on land size, crop type, and resources.*

c) *Prepare a final farm layout plan on chart paper or using digital tools.*

Step 9: acquisition of inputs.

a) *List inputs required for the enterprise.*

b) *Identify possible suppliers.*

c) *Inquire about quality control.*

d) *Calculate estimated costs and prepare an input budget.*

Step 10: implementation and reporting

a) *As a class, prepare a mini-demonstration plot in the school garden using selected crops.*

b) *Apply the farm layout, soil preparation, and input acquisition steps in practice.*

c) *Record all activities and observations in group journals.*

d) *Compile a comprehensive report on the steps of establishing the crop enterprise.*

e) *Present group findings to the class through posters, charts, or oral presentations.*

Guiding questions

- *What is the main purpose of establishing a crop enterprise in the community?*
- *How is land acquired for farming in Uganda and what challenges arise?*
- *Which crops are suitable for the selected site and why?*
- *What factors are considered in a feasibility study for a crop enterprise?*
- *How do site characteristics (soil, water, access) influence enterprise success?*
- *What are the advantages and disadvantages of different land preparation methods?*
- *How is soil tested and which amendments are required for crop production?*
- *What should a farm layout include for effective management?*
- *What inputs are needed and how are they sourced?*
- *How can knowledge of these steps help establish a successful and sustainable crop enterprise in the community?*

9.1 Establishing a crop enterprise

Establishing a crop enterprise involves a series of planned activities aimed at producing crops for food, income, or raw materials. It begins with identifying a suitable crop based on climate, soil, market demand, and available resources. Successful crop enterprise establishment requires careful planning, proper land preparation, selection of quality seeds or planting materials, and timely planting. With good planning and management, a crop enterprise can lead to increased yields, profitability, and sustainable agricultural production.

9.1.1 Identifying the purpose of the enterprise.

Before setting up any crop enterprise, the first and most critical step is to identify and clearly define the purpose of the venture. The purpose shapes all other decisions from the choice of land, the crop, scale of production, investments made, to the marketing strategies adopted.

Common purposes for establishing a crop enterprise.

- **Commercial production**, aim is to grow crops primarily for sale to generate profit. Farmers choose high-demand crops like tomatoes, maize, beans, and watermelons.
- **Subsistence farming,** crops are grown mainly to provide food for the household. Profit-making is secondary. Examples

9.2 Managing a crop enterprise

Class activity 26. Management of a crop enterprise.

Materials needed

- *Farm tools (hoes, pangas, watering cans, pruning knives, spray pumps)*
- *Fertilizers and organic manures*
- *Mulching materials (grass, banana leaves, crop residues)*
- *Plant support materials (stakes, wires, nets)*
- *Seeds and seedlings for demonstration*
- *Protective gear (gloves, masks, gumboots, overalls)*
- *Record sheets/notebooks*

Procedure

Visit a school garden or nearby farm.

Obtain all necessary farm tools and equipment.

Put on protective gear and participate in the following

a) ***Weed control in crops**, demonstrate hand weeding, hoeing, and mulching as weed control methods. Compare chemical (herbicides) vs cultural methods.*

b) ***Disease control in crops**, identify common diseases (fungal, bacterial, viral). Demonstrate control methods.*

c) ***Pest control in crops**, identify common pests. Demonstrate integrated pest management (IPM).*

d) ***Irrigation of crops**, demonstrate watering using cans, furrow irrigation, drip lines, and sprinklers. Discuss when and how often to irrigate.*

e) ***Mulching in crops**, collect mulching materials. Demonstrate application around crops to conserve water and control weeds.*

f) ***Pruning in crops**, show pruning in crops like coffee, tomatoes, or citrus. Remove dead, diseased, or excess branches.*

g) ***Fertilizer application to crops**, demonstrate organic and inorganic fertilizer use. Apply using broadcasting, side dressing, and foliar feeding methods.*

h) ***Provision of support to crops**, demonstrate staking in tomatoes, beans, and passion fruits. Show use of wires, strings, and nets.*

i) ***Training in crops**, demonstrate training in crops like tomatoes, grapes, and cucumbers. Guide plants to grow in desired direction for better yields.*

j) ***Thinning in crops**, demonstrate thinning in crops like maize, millet, and carrots. Show spacing benefits for healthier crops.*

k) ***Gap filling in crop gardens**, replace missing seedlings in a crop stand with healthy seedlings. Maintain uniform crop population.*

l) ***Intercropping in crop gardens**, demonstrate intercropping. Discuss advantages and disadvantages of intercropping.*

m) ***Crop rotation**, design a rotation plan. Discuss how it reduces pests, diseases, and improves soil fertility.*

n) ***Earthing up in crops**, demonstrate earthing up in crops like maize, potatoes, and sugarcane. Show how it supports plants and improves root development.*

o) ***Documentation**, record each practice observed/demonstrated. Note benefits, challenges, and tools used. Compile into a class report.*

3. Inadequate light, it causes etiolation in crops
4. Drought or water stress, causes crops to wilt and die
5. Hail damage which bruises crop parts and destroys crop leaves completely
6. Water logging whereby excess water in soil can cause wilting in some crops due to limited root development and aeration.
7. Air pollution can also cause disease symptoms in crops

Figure 9:2-Common crop diseases

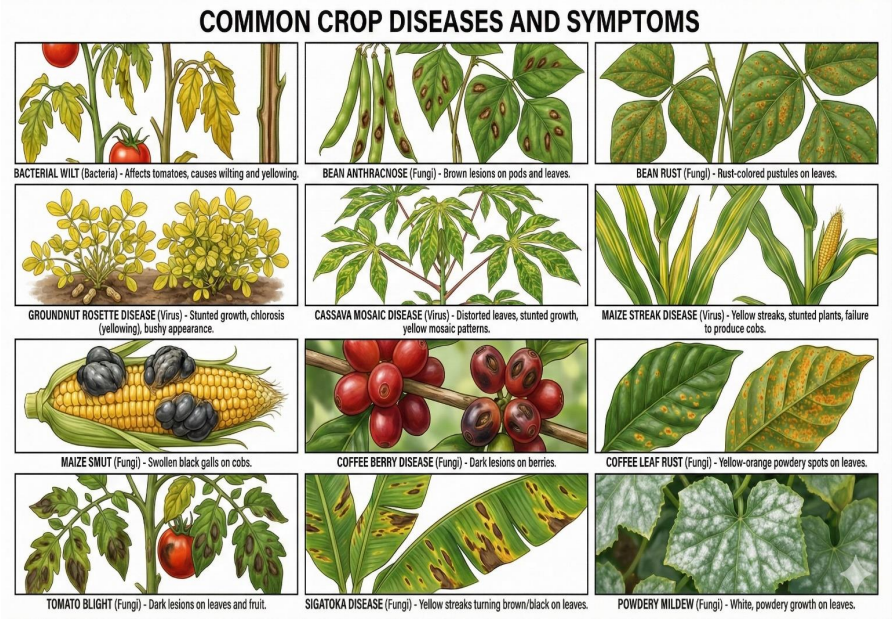


Table 9:8-Common crop diseases, their cause, mode of transmission and control measures.

Disease and causal agent.	**Symptoms.**	**Mode of transmission.**	**Control measures.**
Bacterial wilt. Cause: **bacteria This affects crops like tomatoes**	Wilting of leaves, yellowing of leaves, stem softening and plant death in extreme conditions.	It is soil borne (spreads with in infected soil). It is also spread by contaminated water, tools, seeds and plant debris.	☑ Carry out crop rotation with non-host crops. ☑ Plant resistant crop varieties. ☑ Use clean and sterilised tools. ☑ Ensure proper drainage in the field. ☑ Carry out soil sterilisation and solarization.
Bean anthracnose. Cause: **fungi**	Brown or black lesions on the leaves and branches. Pink or black spots on pods.	Splashes of rain water. Contact with infect plant parts. Planting infected seeds.	☑ Use of clean seeds. ☑ Seed dressing with appropriate fungicide before planting. ☑ Crop rotation with non-leguminous plants.

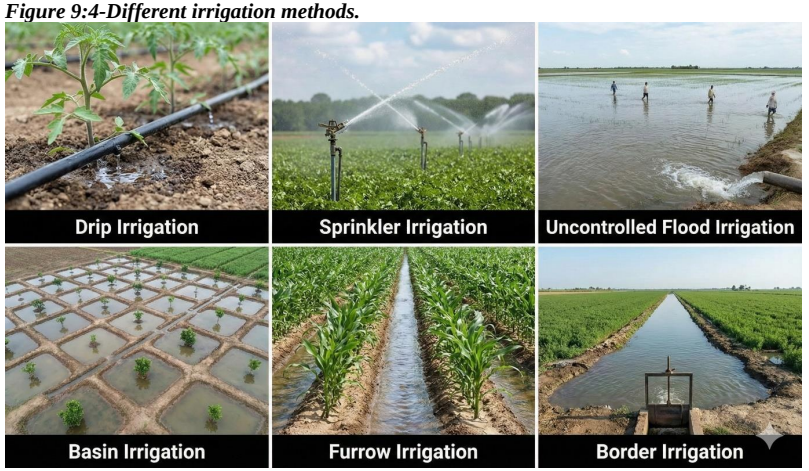
during hot seasons, protecting sensitive crops.

- ❖ **To apply fertilizers and chemicals,** through **fertigation** (applying fertilizers with irrigation water), nutrients can be efficiently supplied to crops, enhancing growth.
- ❖ **To Enhance livelihoods and food security,** irrigation ensures continuous agricultural production, improving rural livelihoods and ensuring national food security.

Methods of irrigation.

The methods of irrigation can be broadly grouped into surface, subsurface and aerial methods.

Figure 9:4-Different irrigation methods.



Factors considered before choosing an irrigation method

- **Water supply**, the quantity and quality of available water must match the system requirements, e.g. flooding needs large volumes while drip needs little but clean water.
- **Labour availability**, the system chosen should match the amount of skilled or unskilled labour available for its operation and maintenance. For example, areal methods require much skills compared to surface methods.
- **Capital outlay**, some systems like sprinklers and drip are expensive to install, while furrows and flooding require less investment.
- **Economic viability**, expensive irrigation systems are only justifiable if the crop enterprise can generate enough returns to cover costs.
- **Soil type**, clay soils retain water longer and suit surface methods, while sandy soils with high infiltration are better for drip or sprinkler systems.
- **Gradient of the land**, flat or gently sloping land suits surface methods like flooding and borders, while steep or uneven land suits sprinklers or drip systems.
- **Type of crop grown**, water-demanding crops like rice require flooding, while high-value crops like vegetables and fruits are best with drip or sprinkler irrigation.
- **Climatic conditions**, hot and dry areas with strong winds suit drip systems (less evaporation), while cooler regions can use sprinklers effectively.

Figure 9:5-Methods of providing support to crops

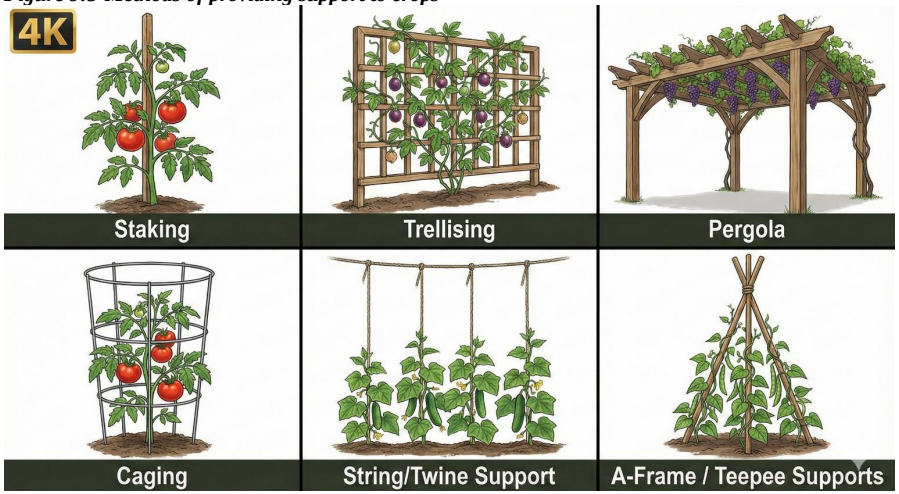


Table 9:12-Methods of providing support to plants, with their descriptions, examples of crops, advantages, and disadvantages.

Method	Description	Examples of crops	Advantages.	Disadvantages
Staking	A single pole (wood, bamboo, or metal) is fixed into the soil, and the plant stem is tied to it to stand upright.	Tomatoes, beans, peppers, yams, eggplants.	♦ Prevents falling and lodging. ♦ Fruits are kept clean and off soil. ♦ It allows easy harvesting. ♦ It improves airflow and reduces disease. ♦ It also saves space.	♦ It is labor-intensive to use. ♦ Stakes rot or break requiring replacement. ♦ It is costly for large fields. ♦ Roots may be injured when inserting stakes. ♦ It requires regular tying.
Trellising	This involves a framework of poles, wires, or strings arranged vertically or horizontally for plants to climb.	Passion fruits, grapes, cucumbers, climbing beans.	✓ It supports heavy vines and fruits. ✓ It allows production of high yields per area. ✓ It allows for better fruit quality. ✓ It allows easier pruning and harvesting.	✓ It is expensive to construct and needs skilled labor. ✓ It requires frequent repair. ✓ It can be damaged by wind. ✓ It is not suitable for very short-term crops.
Pergola	This is a roof-like overhead structure made of poles and wires where vines spread and hang down.	Passion fruit, grapes, gourds, cucumbers.	➔ It allows maximum sunlight use. ➔ It holds heavy loads of vines. ➔ It allows easy harvesting of fruits under the canopy.	➢ It is very costly to install. ➢ It requires skilled labour to design and strong materials to install. ➢ It needs a high level maintenance.

10 CROP HARVESTING AND HANDLING

Competency: The learner harvests optimal quality crop products for the market.

10.1 Pre-harvest practices in crop production.

Class activity 27. Pre-harvest practices done slightly before harvesting.

Materials needed

- *Crop samples (maize, beans, bananas, coffee, cassava, Irish potatoes, mangoes)*
- *Tools (sickles, hoes, pangas, pruning knives, harvesting baskets)*
- *Drying materials (mats, tarpaulins)*
- *Mulching materials, watering cans (for demonstrations)*
- *Notebooks and pens*

Procedure

1. ***Introduction**, discuss as a class what "pre-harvest practices done slightly before harvesting" means. Brainstorm why these practices are important for crop quality and market readiness.*
2. ***Observation and demonstration**, put on protective gear, visit a school garden, community farm, or use crop samples to observe and demonstrate practices in different crops.*
   a) ***cereals (maize, rice, millet, sorghum, wheat),** carry out bird scaring, pest/disease monitoring, irrigation withdrawal to allow drying, mapping of routes, and field drying.*
   b) ***beverage crops (coffee, tea, cocoa),** carry out final pruning to ease harvesting, shade regulation, removing diseased fruits/cherries, mulching to conserve moisture.*
   c) ***legumes (beans, groundnuts, soybeans, cowpeas),** practice bird scaring, weeding to ease harvesting, field drying on ridges before collection among other practices.*
   d) ***fruit crops (bananas, mangoes, citrus, papaya, pineapples),** practice bagging of fruits to encourage proper development, irrigation reduction to improve sweetness, thinning of excess fruits for better quality, briefing of workers and removal of diseased fruits.*
   e) ***root tubers (cassava, carrots, sweet potatoes),** withholding irrigation to prevent rotting, gentle earthing up for better tuber development, pest/disease checks, clearing weeds for easy harvesting among other practices.*
   f) ***stem tubers (Irish potatoes),** carry out withholding irrigation, cutting potato haulms 1–2 weeks before harvest to harden tuber skins, pest and disease control, final earthing up.*
3. ***Recording and analysis,** groups record the specific pre-harvest practices for their assigned crop type. Note the reasons and benefits of each practice.*
4. ***Class presentations**, groups present findings, supported by practical demos (e.g., showing bagged fruits, uprooted beans for field drying, cut potato haulms).*
5. ***Discussion**, compare practices across different crop categories. Discuss how these practices reduce losses, improve quality, and ensure storability.*

Guiding questions

- *What practices are carried out slightly before harvesting each crop type?*
- *Why is irrigation stopped before harvesting some crops?*

harvesting ensures the farmer gains the highest possible yield and income per hectare.

g. ***To comply with market and processing requirements,*** some crops are harvested slightly early or at a specific stage to meet processing or export standards. E.g., mangoes for export may be picked while green but mature. Tea leaves are plucked at the "two leaves and a bud" stage for quality processing.

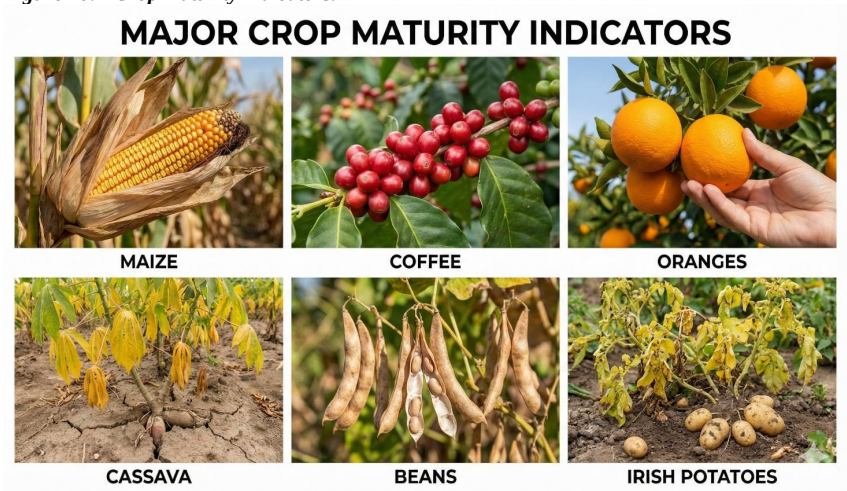
h. ***To ensure seed viability (for seed crops),*** in crops grown for seed production, timely harvesting ensures that the seeds are fully mature**,** viable for germination and free from field infections or weather damage**.** Cereal crops are harvested when the grains are hard and dry.

NB. Before harvesting of crops or crop produce, crops must have obtained physiological maturity. This can be assessed by observing some signs or new characteristics known as ***maturity indicators***

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10.2.3 Maturity indicators.

Figure 10:1-Crop maturity indicators.



Maturity indicators refer to the observable signs, characteristics, or measurable traits that show a crop or its produce has reached the commercial maturity or optimum stage of development for harvesting. These indicators vary by crop type and may be physical, chemical, physiological, or visual, helping the farmer determine the right time to harvest for maximum quality and yield.

Examples of common maturity indicators.

- ✧ Change in color for example green to yellow in bananas.
- ✧ Drying of plant parts for example yellowing of leaves in cereals.
- ✧ Firmness or texture for example softness in mangoes.
- ✧ Size and shape for example full pod size in beans.
- ✧ Sound when tapped for example hollow sound in watermelons.
- ✧ Falling of mature produce.

NB. After crops have attained physiological maturity or commercial maturity assessed through observing maturity indicator. They are regarded ready for harvesting. Harvesting can be done using different techniques. Different crops have different ***harvesting techniques***.

10.2.4 Harvesting techniques for different crops.

Figure 10:2-Crop harvesting methods.



A harvesting technique refers to the specific method or procedure used to collect mature crop produce from the field. It involves the tools, skills, timing, and steps applied to detach or gather crops efficiently, safely, and with minimal damage to both the produce and the plant.

Key aspects of a harvesting technique,

- *Method used* such as hand-picking, cutting, digging, shaking, or mechanized methods.
- *Tool or equipment* involved such as sickles, knives, hoes, harvesters, or threshers.
- *Skill or practice* applied such as careful handling to avoid bruising, and choosing correct maturity stage.
- *Purpose* to ensure maximum yield, maintain quality, and reduce post-harvest losses.

Harvesting techniques for different crops and their maturity indicators.

10.2.5 Cereals. (examples: maize, rice, sorghum, millet, wheat)

Table 10:1-Maturity indicators of cereal crops.

Cereal crop	Maturity indicators.
Maize	Cobs feel firm and full. Husks (covering the cobs) turn dry and papery. Kernels or grains become hard when pressed. Leaves become yellow or brown. Ears begin to dry. No milky juice in pressed grain. When a black layer forms on the grain.
Rice	Panicles (ears) turn golden-yellow. Grains become hard and may not be easily dented by nail. Lower leaves dry and brown. Whole plant looks dry, bends. No milky juice in pressed grain.
Sorghum	Panicles or ears turns brown or reddish. Grains are hard and shiny. Leaves dry from base upward. No milky juice in pressed grain. When a black layer forms at the grain base.
Millet	Heads begin to drop. Grains are firm and dry. Plant color turns to yellow. No juice from pressed grains.

11 CROP AND CROP PRODUCT MARKETING

Competency: The learner markets crop and crop products for profit.

Class activity 29. Marketing and market analysis in crop enterprises

How can we effectively market crop products, manage cash flows, and stabilize market prices to support farmers in our community?

Materials needed

- *Questionnaires, interview guides, checklists for surveys*
- *Crop product samples (maize, beans, coffee, tomatoes, bananas, etc.)*
- *Packaging materials (sacks, cartons, bottles, labels)*
- *Financial templates (cash flow sheets, calculators, spreadsheets if available)*
- *Graph papers, rulers, markers for Cobweb diagrams*
- *Reference documents (local market bulletins, UNBS standards, export guidelines)*
- *Chart papers, flip charts, manila cards, markers for presentations*

Procedure

Phase 1. Marketing functions and regulations.

a) *Form groups and select one crop or crop product (e.g., maize flour, beans, coffee, bananas).*

b) *Conduct a market survey with farmers, traders, or cooperatives for a given product.*

c) *Investigate, marketing functions and their local regulatory requirements and international standards regarding given product.*

d) *Prepare a leaflet or report showing how marketing functions are applied and which regulations govern them.*

e) *Present findings in class, with demonstrations (e.g., mock packaging or labeling).*

Phase 2. Cash flow management.

a) *Using the selected crop, research real or estimated costs of production, transport, and marketing.*

b) *Prepare a cash flow statement showing:*

- ✓ *Inflows (sales, subsidies, credit).*
- ✓ *Outflows (inputs, labor, storage, transport, marketing).*

c) *Analyze whether the enterprise is sustainable (profit vs loss).*

d) *Suggest how surpluses can be reinvested and how deficits can be addressed (e.g., loans, collective marketing).*

e) *Present results using tables, charts, or financial summaries.*

Phase 3. Applying the cobweb theorem.

a) *Teacher introduces the Cobweb Theorem.*

b) *Groups research and collect market price data for their crop (weekly or seasonal fluctuations).*

c) *Predict future market fluctuations based on observed prices.*

d) *Propose strategies to stabilize prices of products.*

Present findings with posters, graphs, and stabilization strategies.

Guiding questions

- *What marketing functions are required for crops and how are they applied?*
- *What regulations (local and international) affect crop marketing?*
- *How can cash flows be recorded and managed in a crop enterprise?*
- *What challenges do farmers face in managing money from crop sales?*
- *How do supply and demand fluctuations affect crop prices?*
- *How does the Cobweb Theorem explain these fluctuations?*

The Cob Web theory can be applied in crop and crop product marketing in the following ways.

- ✓ **Influence of previous prices on farmers' decisions,** farmers often decide what to plant depending on last season's prices. If beans fetched high prices last year, many farmers will increase bean production this year. This sudden increase in supply may flood the market, reducing prices when the crop reaches the market.
- ✓ **Time lag between planting and harvesting,** since crops take time to grow, the market cannot adjust supply immediately. This delay means that even if current prices fall, farmers have already planted, leading to an imbalance between supply and demand when harvest comes.
- ✓ **Market fluctuations in crop prices,** the alternating cycle of **high prices–low supply** followed by **low prices–high supply** is common in crop marketing. For example, when maize prices rise sharply in one season, farmers rush to plant more, leading to oversupply and low prices in the next season.
- ✓ **Predicting market behavior,** marketers, cooperatives, and governments can use the Cobweb Theorem to predict fluctuations in supply and prices**.** For instance, after observing high tomato prices, they can anticipate oversupply in the following season and prepare by creating storage facilities, processing, or advising farmers to diversify.
- ✓ **Guiding marketing strategies,** the theorem helps in planning marketing interventions such as:
  - ❖ Organizing storage to obtain more profits.
  - ❖ Processing crops into value-added products to absorb excess supply.
  - ❖ Diversifying crops to avoid price crashes in one commodity.
  - ❖ Timing market entry for example selling crops when prices are favorable rather than during gluts.

Trial items 11:1-About crop and crop product marketing.

Item one.

A crop farmers' association has been supplying grain to traders and institutions. However, buyers frequently complain about poor quality, inconsistent standards, and failure to meet market requirements. Some contracts have been cancelled, and the association's income has declined.

Task.

Advise the farmers' association on how proper marketing functions and adherence to market regulations can improve their market access and profitability.

Item two

A farmer grows vegetables for sale in nearby towns. He often produces large quantities based on what other farmers are planting. During some seasons, he earns good money, but in other seasons, he sells at very low prices and struggles to recover production costs. He does not keep financial records and rarely seeks information from buyers.

Task.

Explain how market survey and proper cash-flow management can help the farmer achieve stable income from his crop enterprise.

Item three.

In a certain district, farmers tend to increase production of crops that had high prices in the previous season. This often leads to oversupply and price drops. In the following season, production declines and prices rise again. This cycle has continued for several years.

Task.

Write an essay discussing how the cobweb theorem explains this situation and how farmers can use this knowledge to make better production and marketing decisions.